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Tuning the Digital Soundscape: The Intersection of Audio Signal Processing, Neural Networks, and Fog Computing in the Internet of Sound Things (IoST)

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Abstract: Sounds are a valuable source of data, and even the absence of sound conveys information. Recent technological developments in sound processing, such as IBM Acoustic Insights, extract meaningful information from the sounds generated by machines. These advancements, coupled with the rapid growth of Internet of Things (IoT) applications, have given rise to the Internet of Sound Things (IoST) paradigm. IoST networks involve computing devices attached to physical objects and focus on creating, recognizing, processing, and interpreting musical and non-musical audio content in distributed environments using machine learning (ML) and deep learning (DL) methods. However, challenges such as audio signal interference, noise, high latency, and the computational complexity of real-time audio processing persist. Additionally, DL approaches face issues that are particularly relevant to dynamic environments like IoST, including concept drift, overfitting, underfitting, and catastrophic forgetting. These issues affect model performance and reliability, necessitating advanced techniques to achieve long-term accuracy and generalization. This review examines the potential of fog computing and DL to synergistically co-optimize current IoST architectures and process real-time audio streams. The review highlights important technical intersections, such as reducing latency through fog-based processing, edge-side processing, and adaptive learning solutions. This review of current limitations and evolving strategies will guide future research toward scalable, intelligent, and latency-resilient IoT systems.

Keywords: Audio Processing, Deep Learning, Fog Computing, Internet of Sound Things, Data Stream, Real Time

1. INTRODUCTION

Sound is a disturbance of particles in the surrounding medium caused by movements of a physical object [1]. This disturbance travels as a mechanical wave through a medium. A microphone converts it into an analog signal, which an analog-to-digital converter (ADC) transforms into digital form, called "audio." Sound contains a lot of information that can reveal the current location of objects. The absence of sound may indicate that an object is not moving, or that there are limitations in the transmitting medium or the receiver. Recently, the ability to collect, save, share, and receive sound has become much simpler due to smartphones, voice assistants, and audio-enabled devices. The audio equipment industry is expected to grow from 34 billion in 2022 to 42 billion in 2026 (5.4% CAGR) [2]. In addition, the audio digital signal processing industry is expected to increase in size from 9.618 billion USD in

2020 to 17.7 billion USD by 2026 (9.3% compound annual growth rate (CAGR)) [3].

This revolution in sound processing has resulted in a focused Internet of Things approach known as the Internet of Sound Things (IoST). While this concept is often referred to as the Internet of Sound (IoS), this study adopts the term Internet of Sound Things (IoST) to distinguish it from Apple's operating system (iOS). A formal definition of IoST is: "The set of sound things, protocols, network infrastructure, and information related to sound to enable applications for the communication of information related to sound in the physical/digital domain. The sound thing is a networked computing device (equipped with actuators and/or sensors) with the ability to capture, process, produce, or exchange sound or sound or its meaningful information" [4].

However, realizing the full potential of IoST presents significant challenges. Although advances in sound processing technology enable the extraction of more information from audio data, processing this information remains complex. Audio signals, which are non-stationary, meaning that their properties change over time, which complicates processing [5]. The combination of this characteristic with variations in signal quality and the presence of background noise makes audio data particularly challenging to process compared to other data types. While audio processing applications or algorithms may successfully perform basic functions, they often lack optimal performance. In order to achieve the full potential of the Internet of Sound Technologies (IoST) in different industries, these challenges must be addressed.

The challenges of audio processing in IoST make the task complicated, hence the use of deep learning (DL) methods. DL refers to an artificial neural network (NN) with multiple hidden layers. These methods excel at detecting trends and extracting features from complex, noisy signals, such as audio. Learning features from data in DL is beneficial because it reduces noise levels and denoises data, resulting in improved signal classification and higher accuracy in audio processing applications. However, applying DL to audio processing is not without complications. It raises questions about the model's interpretability, robustness to certain conditions, and high computational requirements [6]. These challenges are compounded when working with audio data, since this kind of data consumes a lot of storage space and bandwidth compared to other types of data. For example, a three-minute audio recording that contains stereo, 16-bit, 48-kHz data needs roughly 34.5MB, while a couple of typewritten pages will take up only 1 to 2 kilobytes [7], [8].

Cloud computing is critical in meeting the computational cost and storage demands of sophisticated audio processing in IoST applications. Through the use of hardware and software virtualization, cloud computing enables the dynamic sharing and allocation of resources [9] that supports audio processing algorithms.

Although cloud computing offers tremendous resources for IoT applications, escalating customer expectations for real-time responses reveal some constraints, especially regarding latency. These challenges are further complicated by continuous data streams over networks. To overcome these constraints, the ancillary model of fog computing has emerged, migrating some computational workloads from the cloud to the edge of the network. This approach considerably reduces latency. This approach brings processing closer to the data source, offering a more efficient solution for time-sensitive audio applications within the IoT.

Although IoST is still in its infancy, it has already received significant attention from researchers, resulting in several comprehensive review studies. [4] provides an

overview of the fundamental elements of IoST systems, including their architectural design, protocols, and security challenges. The authors emphasize the need for ongoing research to improve the security of IoST systems. [10] focus on developing and implementing IoT systems to monitor sound levels, specifically in environments that are crucial to the health and safety of senior citizens and individuals with special needs. The authors used 6LoWPAN, a low-power wireless network protocol, to provide efficient data transmission with minimal loss, providing real-time environmental monitoring and responses to hazardous noise levels. Another aspect of IoST is covered in [11], where the question of acoustic attacks on the IoT infrastructure is studied.

There are different technical issues in IoST that overlap: audio processing, DL, fog computing, strict timing, and handling dynamic data streams, so the aim of the research is to study the following:

- What challenges does audio stream processing present in IoST?
- How can DL methods support IoST applications taking into account the stream processing requirements?
- How does fog computing aid delay-sensitive applications in IoST?

The main contribution of this review is to provide the basis for future in-depth research that meets the IoST specific requirements and overcomes its overleaped challenges.

The rest of the paper is organized as follows: Section 2 reviews the evolution of audio processing and the application of NN in this field. Section 3 examines the use of sound to sense the environment and highlights one of the key IoST applications. Section 4 analyzes the fog computing architecture that supports the deployment and functionality of IoST applications. Sections 5 and 6 address key problems and complexities in audio signal processing, particularly the complexity of audio signals themselves and NN approaches. Finally, Section 7 presents the conclusions of the review and outlines future research directions.

2. THE EVOLUTION OF AUDIO PROCESSING: FROM THE PHONOGRAPH TO NEURAL NETWORKS

A. Early Developments

Since its inception in the mid-19th century, the field of audio processing has undergone significant advances. This section explores the key milestones in the development of audio technology, from the early days of mechanical devices to the modern era of digital systems.

The history of audio processing can be traced back to 1857 when Edouard-Leon Scott de Martinville received a French patent for the phonautograph [12]. This device is the first attempt at creating a visual representation of sound waves. The phonautograph used a stylus vibrated

by sound waves to etch patterns onto a moving surface, effectively creating a visual analog of audio signals. In 1878, Thomas Edison made a groundbreaking invention with the phonograph [13]. A major advancement in audio technology was the phonograph's ability to record sounds onto tinfoil cylinders and play them back.

B. Theoretical Foundations and the Analog Era

While the field was still progressing, the theoretical foundations were established that would be put to great use later on. In 1928, Harry Nyquist published what would later be called the "Nyquist Theorem" [14]. The Nyquist theorem states that, to avoid aliasing and ensure the correct representation of the original signal, we must reconstruct a continuous signal from its sampled form based on sampling rate, which is at least double the highest frequency constituent of the signal. The sampling rate is the number of samples taken from an analog signal per second when it is converted to a digital format.

Various analog audio processing devices came into being in the mid-20th century. One such device was the Pultec EQP 1, an analog equalizer introduced in 1950 [15]. This product made it possible to precisely shape and control certain frequency bands. This made it one of the first modern audio processing devices capable of controlling individual frequency bands during recording and playback. The 1960s saw significant advances in recording and playback technology. In 1963, the Philips Company presented the compact cassette tape format and offered licenses worldwide [16]. This format became a standard for portable audio recording and playback for decades. In 1965, Dolby Laboratories introduced the Dolby Type A noise reduction system [17]. Dolby A was designed to reduce tape hiss, a common problem in analog recordings. It employed a sophisticated approach. During recording, high frequencies were pre-emphasized to increase the amplitude in the relative to the hiss of the tape. Then, during playback, a corresponding de-emphasis filter scaled the high frequencies back down to their original levels with respect to other signals that are recorded. The pre-emphasis filter was amplitude-sensitive, emphasizing only soft high-frequency sounds. This prevented excessive emphasis on high frequencies, which could lead to distortion [18].

C. Digital Revolution and Neural Networks

The rapid invention of electronic instruments, processing methods, and communication technologies in the 20th century accelerated the development of new forms of audio processing. As technology advanced, the field began to embrace digital techniques and, eventually, artificial intelligence (AI). In 1943, the neurophysiologist (Warren McCulloch) and the mathematician (Walter Pitts) presented the first mathematical model of an NN [19], while in 1954, (Belmont Farley) and (Wesley Clark) managed to running the first simple NN on a digital computer [20]. However, it would take several decades before NN would be applied effectively to audio processing. NN are advanced, non-linear

statistical data processing frameworks that have potential applications in decision-making processes [21]. They can interpret complex relationships between independent and dependent variables, as well as input and output variables, and recognize patterns within data sets. Audio data is often complex, especially when multiple signals are present at the same time and within the same frequency range. Separating them can be challenging because there is no general method for resolving sound components [22].

D. Neural Networks in Audio Processing: Innovative Insights

One significant early application of NN to audio processing was presented in [23]. The authors proposed a time-delay neural network (TDNN) for phoneme recognition. TDNN was trained and evaluated on a dataset of speech recordings. In TDNN the basic units can compare the current input to the history of events. In doing this, TDNN can learn the relationships between the temporal acoustic characteristics of a specific time offset. For example, to distinguish between the phonemes "a" and "e", "a" and "i", TDNN can learn the acoustic features that occur immediately before and after the phoneme "a" (because they help to recognize "a" within other phonemes). Hidden Markov Models (HMM) achieved 93.7% recognition rate; the TDNN presented in [23] achieved 98.5%.

The authors of [24] introduced a novel method for effective speech recognition using a hybrid system of HMM and NN, where the NN acted as a vector quantizer (VQ) to preprocess speech input. Rather than using a k -means clustering vector quantizer, as in traditional systems, NN was trained with an unsupervised learning algorithm, maximizing mutual information (MMI). The NN underwent this training method and outputs probabilistic values that fed the HMM. The resulting system reduces recognition errors by aligning the training of both the NN and the HMM, resulting in a more robust system than one based on HMM only models.

Furthermore, the authors of [25] present a novel approach to scene classification using audio features without the need for visual features. The data set for the NN training contained audio clips from different TV programs. The audio features extracted by the authors include volume distribution (mean, standard deviation, silence ratio, and dynamic volume range), pitch contour, and frequency features (frequency centroid, bandwidth, and subband energy ratios). To evaluate these features, they calculated intra-class and inter-class scattering matrices to assess their discriminative capability. An NN classifier with a one-class-in-one-network (OCON) structure was used for scene classification, where each subnet specialized in recognizing one specific scene class. According to [25], NN, specifically feedforward types, have been effective in pattern classification tasks despite some challenges. For example, conventional multi-layer perceptron (MLP) adopts an all-class-in-one-network (ACON) structure that requires more hidden units due to

the complexity of catering to all classes. Therefore, [25] proposed a simpler approach using the OCON structure, where each subnet identifies a single class [26]. Training each subnet individually with the backpropagation algorithm ensures that the output will be close to 1 for relevant inputs and close to 0 for the rest.

Until 2003, traditional gender identification systems often struggled under variable speech conditions, such as background noise and language differences. To address these challenges, the authors of [27] proposed an approach that maintains high accuracy in gender classification, regardless of the quality or characteristics of the speech data. The authors used a general audio classification approach that uses long-term spectral features, deviating from traditional pitch-based or phoneme-dependent methods. The signal is segmented into nonoverlapping 1-second windows. In each window, the model calculates the variance and mean of the Mel Frequency Spectral Coefficient (MFSC) vectors. The authors experimentally showed that the MFSC features slightly outperform Mel Frequency Cepstral Coefficients (MFCC) in their model. MFSC and MFCC are audio features which capture most of the spectral features of sound to allow speech and sound classification, based on the perception of the human auditory system. However, the classifier does not capture frequencies themselves, but rather the relationships between them. The model was language independent, robust to loudness and audio compression. It also achieved high accuracy.

In [28], a novel model based on recurrent neural networks (RNN) was introduced for automatic speech recognition (ASR). ASR systems face the challenge of dealing with noise that has a significant impact on speech recognition accuracy. Deep-recurrent autoencoder NN was proposed to denoise input speech features and improve ASR performance in the presence of noise. In contrast to conventional approaches that depend on manually designed features or a particular noise model, this method learns general denoising directly from the data, without constraining the noise model. The model architecture consisted of many hidden layers and recurrent connections to model speech and noise dependencies over time. The deep recurrent denoising autoencoder (DRDAE) minimizes the variation between the predicted and the actual clean speech features. The success is attributed to its recurrent nature, which explains the correlation across time steps. The evaluation performed on the Aurora2 dataset yielded substantial reductions in word error rates compared to traditional noise reduction systems.

In [29] an approach based on DL is introduced to combine audio and visual information to improve the clarity and quality of noisy speech signals. The authors present a multimodal, deep convolutional neural network (CNN) that is trained to independently fuse two streams of audio and visual data and output the combined result. Extracting audio and visual features includes identifying lip movements. These features are run through separate CNNs and merged

into a joint network, enabling the model to learn audio and visual cues and use them to enhance corrupted speech signals. The model was trained end-to-end with a multitask learning framework. Not only did the system improve the word recognition rate considerably, it also performed well in noisy environments.

A new RNN based Long Short-Term Memory (LSTM) for the automatic detection of abnormal heart sounds is described in [30]. This model is designed to capture the temporal dependence of sequential structured data. The authors preprocessed the data by extracting important features using MFCC and isolating noise using a bandpass filter. They also employed LSTM layers to learn and memorize long-term relationships between normal/abnormal heartbeats. The final layer uses a softmax function to classify heartbeats into one of eight abnormal types. The authors tested the performance of the model and compared it with other classifiers such as MLP, Decision Trees (RF), Naive Bayes (NB), Support Vector Machine (SVM) and K-Nearest Neighbors (KNN). The model showed high and improved classification accuracy on two benchmark datasets: 99.71% on the Pascal Challenge and 98.7% on the PhysioNet 2017 dataset.

The authors of [31] proposed a flexible framework called NeuProNet (Neural Profiling Network). Unlike traditional audio classification models, which treat each audio sample independently, NeuProNet groups samples with shared characteristics into candidate sets. It then applies a contrastive learning mechanism to learn personalized, high-level features. This approach enables the model to capture shared attributes across grouped data and distinguish relevant features from noise. The NeuProNet framework can be integrated into existing backbone architectures, such as ResNet, EfficientNet, or Transformer models. It is trained in an end-to-end manner, which ensures that profile and classification features are optimized simultaneously. This leads to improved accuracy and generalization in sound classification tasks.

Advancements in audio processing, ranging from basic gadgets to complex DL methods, have increased the usefulness and complexity of sound data systems. The NN-based techniques discussed in this section, such as TDNN and LSTM, have improved the accuracy of audio feature extraction and classification, but they typically require extensive preprocessing and a broad range of audio feature extraction prior to training. Additionally, signals in the IoST are often non-stationary, so relying on these methods can be challenging when tasks must be performed in near real time. However, sights and sounds surround us, and using them well can create a much better living environment. Thus, despite the challenges, the superior accuracy of NN models is driving the adoption of more NN applications in the IoST environment, such as environmental monitoring.

3. UTILIZING SOUNDSCAPE FOR ENVIRONMENTAL SENSING

Soundscape concept was introduced by R. Murray Schafer in the 1977 book 'The Tuning of the World' [32].

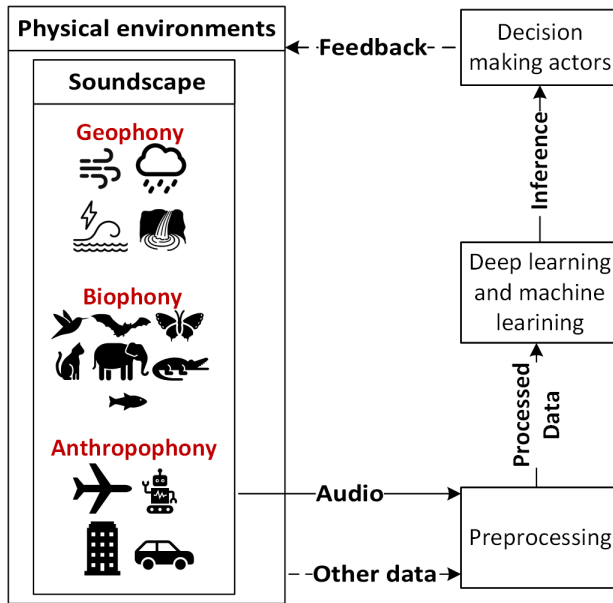


Figure 1. The general schema for environmental sound analysis

Schafer defined a soundscape as the collection of sounds that humans hear in a given environment. According to [33], developments in the field of recording and data analysis require closer and more detailed research on the complex sources and environmental contexts of soundscape acoustics to better understand this phenomenon. A soundscape consists of three primary acoustic sources biophony, geophony, and anthropophony which are sometimes independent and sometimes interfere with each other in different ways. [33]. Environmental sound analysis has the potential to transform how sound affects human life. It also provides opportunities to develop methods that optimize the performance and efficiency of objects such as turbines, vehicles, and medical devices. Even silence can serve as an indicator of physical state.

Figure 1 represents a general abstracted schema for an environmental sound analysis system that uses DL and ML for decision making where:

- **The Physical Environments and Soundscape Composition:** Soundscape is the foundation of the framework comprised of different Physical Environments. This soundscape is classified into three primary categories: biophony, geophony, and anthropophony. Geophony consists of non-biological natural sounds such as wind, rain, ocean waves, and waterfalls, and represents the natural ambience of an environment. Biophony consists of sounds of biological origin, such as sounds made by birds, insects, and mammals. This can indicate the biodiversity and ecological dynamics of a habitat. Anthropophony is defined as human-generated sounds from activities such as traffic, industrial activity, and machinery that typically indicate

anthropogenic impacts on natural settings.

- **Data flow:** Environmental audio signals along with other contextual data (e.g. meteorological conditions, timestamps, and location metadata (LMD)) are collected from the environment for preprocessing. This multimodal approach guarantees a global understanding of the context for better sound analysis.
- **Preprocessing** takes these data and refines them by filtering out (signal) noise, normalizing signals, and reformatting the audio into a representation that is most useful for the analysis, spectrograms, or MFCC. During this step, input formats are standardized and meaningful features can be extracted, making the data ready for machine learning analysis.
- **DL and ML analysis:** The models are used for tasks such as pattern recognition, anomaly detection, and environmental sound classification. Advanced architectures, such as CNNs, RNNs, etc, could be used to learn spatial and temporal features from the audio data. During this stage, data are transformed into output distributions, which produce classifications, probabilistic predictions, or other forms of decision support metrics.
- **Inference and Decision-Making Process:** Decision-making actors such as scientists, policymakers, or automated systems generate insights using the ML output which is interpreted as Inference.
- **Feedback to the physical environment** from decision-making actors is a feature of the framework. Adaptive management of feedback allows the system to adjust its analysis parameters or data collection methods corresponding to the actors' decisions. For example, if there is a need to implement a policy change to reduce noise pollution in a certain place, the system can keep track of changes in the soundscape and make an analysis accordingly. Feedback in this case strengthens the system's responsiveness to change and makes adjustments that support sustainable environmental practices and real-time monitoring.

There is a lot of active research in this field. For example, [34] developed a system that classifies audio signals in security and military settings, such as gunshots and explosions. The system supports an audio source localization system that can respond effectively to such sounds. The method employs frequency-based audio feature extraction, using techniques such as Power Spectral Density and Fourier Transforms to analyze audio signals. A feed-forward NN, specifically an MLP with three layers, was trained using supervised learning to classify sounds such as heavy weapons, light weapons, and explosions. The system was implemented using MATLAB NN Toolbox, with the Levenberg-Marquardt algorithm optimizing the model, and mean squared error (MSE) was used as a performance

metric. Field tests were conducted, where microphone units with GPS, 2.4 GHz radio modules and parabolic antennas were placed around a secure area. The microphones were set up in a non-linear arrangement to capture and transmit audio signals and their locations. The decentralized evaluation unit was strategically placed near the base station to process the data.

The authors of [35] created an automatic acoustic novelty detection system, using approximately three hours of home environment recordings. The system identified acoustic signals that deviated from the normal reference data on which the system was initially trained. The method leverages auditory spectral features in conjunction with DAEs integrated with Bidirectional Long Short-Term Memory (BLSTM) networks and focuses on the reconstruction error produced when the DAE attempts to process new or anomalous sounds. The combination of memory-based processing and autoencoding enables acoustic novelty detection without the need for labeled datasets. This distinguishes the method from existing supervised or semi-supervised techniques.

The authors of [36] proposed a novel hybrid NN architecture for the identification of vehicle types from audio data recorded from roadside microphones. This hybrid architecture improves the NN classifier by integrating LSTM units with CNNs. The LSTM, a specialized type of RNN, is utilized to address the issues of vanishing and exploding gradients during long-sequence training. The authors incorporated a novel feature set into the hybrid NN to accurately classify different types of vehicle. They identified three key audio features essential for vehicle identification: MFCC, Pitch Class Profile (PCP), and Short-term Energy E_n .

However, the proliferation of microphones and mobile devices has made environmental sound monitoring increasingly challenging. More sound can now be recorded on a greatly increased and distributed scale across a large number of geographies. The expansion of audio data collection creates additional challenges when transferring audio data across the Internet to processing centers and waiting for results. Complexity issues, such as large file sizes, interference, and noise, require efficient preprocessing and may be triggered locally before main processing on a remote server. These issues have greatly increased the need for advanced computing architecture in recent years to speed up decision-making.

4. FOG COMPUTING AND THE IoST

The collection, analysis and exchange of sound-related data within IoST face strict constraints and challenges. For example, the network infrastructure must ensure millisecond-level latencies to ensure no discernible delays. The challenges such as network bandwidth limitations, limited computational power, and power issues which are prevalent in IoT devices are also issues for IoST. IoST requires the development of specialized analytical tools tailored to sound processing. These tools must handle vast

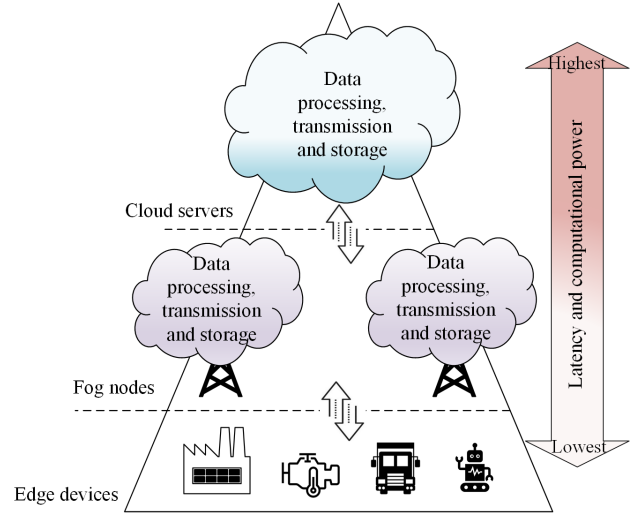


Figure 2. The concept of fog computing

amounts of sound-related data while extracting valuable insights under stringent latency and energy constraints. This presents distinct challenges for NN and real-time signal processing.

Consequently, cloud computing alone cannot fulfill the requirements of time-sensitive applications due to its high latency. Fog computing mitigates this limitation by relocating part of the computational resources from the central server to the network's edge. In critical applications, particularly within industrial settings, the implementation of fog computing for machine monitoring reduces service time, minimizes machine downtime, and improves machine reliability and availability, thus improving overall production efficiency [37]. Figure 2 shows the concept of fog computing in which a hierarchical model is obtained for data processing, transmission, and storage in distributed systems. Edge devices shown in the base are data sources or rapid minimal processing, while fog nodes perform rapid processing with low latency, near the data sources. Cloud Servers tackle heavy data processing, long-term storage, and computation with large power but higher latency at the top. This layered approach makes a trade-off between latency and computational demands while remaining efficient with latency sensitive and data intensive applications. Such an architecture holds tremendous opportunity to create an IoST system that can serve a large number of domains and uses.

We categorize some efforts as IoST applications within the fog computing environment, where some of the processing is done close to data sources, and other complex computation is migrated to remote servers. For example, [38] propose a two-tiered sound recognition platform aimed at monitoring environmental conditions. The platform consists of wireless acoustic sensor network (WASN) nodes that:

- 1) collect sound samples from the environment,

- 2) perform denoising using wavelet techniques,
- 3) extract relevant features,
- 4) apply a Gaussian Mixture Modeling unit to capture the variability of sound classes,
- 5) transmit the results to a central server using wireless communication protocols.

The central server in the platform conducts automatic sound classification using a two-layer NN. The system categorizes sounds into three main classes: anthropogenic, biophysical, and geophysical. These are further divided into nine specific subclasses: anthropogenic sounds include as example airplane, car, and pistol, while bird, dog, and snake represent biophysical sounds; and fire, waterfall, and wind for geophysical.

Additional efforts, such as those provided in [37], focus on fault detection in industrial machinery by analyzing operational sounds. In this approach, machine sounds are classified as normal or abnormal. The authors compared the performance of a remote cloud server with fog computing and found that fog architecture achieved a better response time. They also evaluated the effectiveness of various ML models in performing this classification. Regarding audio features, the authors propose the extraction of Linear Prediction Coefficients (LPC) and MFCC. LPC method is used to model speech production by predicting future speech samples as a linear combination of past samples.

Although NN have proven effective for audio processing and fog computing architecture supports the needs of Internet of Things (IoT) systems by facilitating scalable applications across distributed environments, audio processing still faces unique and significant challenges. The quantity of audio data, the requirement of feature extraction, and the complexity of the audio signal itself present greater challenges than other types of data. Given these complexities, audio processing is a non-trivial task, and further innovation and effort are needed.

5. THE INTRINSIC COMPLEXITIES IN AUDIO SIGNAL PROCESSING

Processing audio streams in IoST is complicated. The challenges stem from several factors, whether ML approaches are used or not.

A. Storage and bandwidth

Unlike text-based data formats such as JSON, which are commonly used in digital communication, audio demands a significantly larger amount of data.

B. Non-stationarity

Time varying components in Audio signals make them inherently non-stationary rendering Algorithm design and adaptive feature extraction necessary [5]. Variability may be introduced due to:

- Source variability, for example, the fundamental frequency or speech rate change over the age of the speaker or with illness are impacting systems like sound security [39],
- Environmental sources such as changing media density, background noise or interference altering amplitude and signal quality,
- Processing variability such as changes in recording or editing quality.

Therefore, a dynamic processing strategy is required for the IoST applications.

C. The Cost of Multi-Domain Processing

Many of the features extracted from audio signals require conversions between the time and frequency domains. Feature extraction can be computationally demanding, particularly in real-time audio processing or when dealing with long audio signals.

D. The Complexity of Audio Stream Processing

A data stream is a sequence of data elements that appear over time [40]. Stream processing performs continuous analysis of incoming data, using algorithms that process unbounded streams to produce an output or respond to a query within a sliding window [41], [42]. Examples of stream processing are applications such as automatic and real-time speech and speaker recognition [43], also real-time sound source localization system [44]. Such applications generate output based only on the current state of sample (stateless processing) and/or recent state of samples (stateful processing). Stateful processing, which tracks historical data from specific sources, is CPU-intensive and requires resources that complex to scale well [45], [46]. To handle such complexity, one may even require a stateful virtualization infrastructure that is able to preserve the state of data and processing under different circumstances, which is critical in some applications such as the Industrial Internet of Things (IIoT) [47], [48]. These issues present real challenges to efficient, real-time audio processing and must be addressed.

The many complications of audio signal processing, such as the need for large storage, non-stationary issues, working with multiple domains, and stream processing, present significant challenges for IoST systems. These challenges demonstrate that a central difficulty is performing real-time actions while maintaining accuracy in unpredictable environments. Additionally, due to non-stationarity and limited bandwidth, adaptive approaches and edge processing are necessary to reduce latency. Nevertheless, running feature extraction methods places a significant load on the system, underscoring the importance of optimized, IoST-targeted approaches in distributed audio programs. While DL methods for audio processing have demonstrated outstanding performance across a range of applications, they face significant challenges when applied to real-time and streaming data applications such as IoST. The next section discusses the most significant of these challenges.

- Source variability, for example, the fundamental frequency or speech rate change over the age of the

6. CHALLENGES OF NEURAL NETWORKS IN THE INTERNET OF SOUND THINGS

A. Concept drift

Non-stationary audio could cause the underlying data features and distributions to change unexpectedly. In ML and DL, this can cause concept drift when unforeseen changes affect the predictive accuracy of the model. Therefore, methods and knowledge acquired after model training may require ongoing updates.

To formalize the concept drift, firstly consider a set of input features $X \in R^p$ with the goal of predicting a target variable $\hat{y} \in R^1$ in regression tasks (or $\hat{y}$ categorical in the classification problems). In these scenarios, at the time of the prediction X is known while $\hat{y}$ is unknown. According to Bayesian Decision Theory, classification problems can be described using $p(\hat{y})$ (the prior probabilities of the classes) and $X|\hat{y}$ (the class-conditional probability density functions) for all classes $\hat{y} = 1, \dots, c$ where c is the number of classes, thus the classification decision for class $\hat{y}$ can be represented as [49]:

$$p(\hat{y} | X) = \frac{p(\hat{y}) p(X | \hat{y})}{p(X)}, \quad p(X) = \sum_{\hat{y}=1}^c p(\hat{y}) p(X | \hat{y}) \quad (1)$$

where $p(\hat{y}|X)$ is the posterior probability of class $\hat{y}$, given the features X .

Figure 3(a) depicts the original distribution example in which three classes of data points, white, black, and gray diamonds, are separated by lines representing the decision boundaries of the classifier, based on the learned rules from the model. The model uses boundaries to separate the points based on their features (X) into their respective targets ($\hat{y}$), for example, to determine which class a point belongs to.

So, formally concept drift between time t_0 and time t_1 can be defined as [49]:

$$\exists X : p_{t_0}(X, \hat{y}) \neq p_{t_1}(X, \hat{y}) \quad (2)$$

where p_{t_0} is the joint distribution at t_0 between X and $\hat{y}$. The prediction may be affected by changing the posterior probabilities of classes $p(\hat{y}|X)$ which are caused by changing the prior probabilities of classes $p(\hat{y})$ and/or class conditional probabilities $p(X|\hat{y})$ [49].

According to [49], [50], [51], the drift can be classified into two main types based on probability or speed. Concept drifts in terms of probability include several types and as follows:

- Figure 3(b) provides an example of a virtual concept drift involving a change in the class conditional probability $p(X|\hat{y})$ without affecting $p(\hat{y}|X)$, impacting the data distribution within the same class without altering the decision boundary, potentially leading to performance issues.
- Figure 3(c) provides an example of a real concept

drift where the boundaries between the three classes have moved due to a change in the relationship between the input features (X) and the target variable ($\hat{y}$). Although the data keep the same spatial distribution, the learned decision rules from the original distribution are no longer valid.

- Figure 3(d) illustrates class imbalance drift, denotes a change in the prior probabilities $p(\hat{y})$, causes the proportion of some classes to change, potentially leading to one class dominating the dataset.
- Figure 3(e)) shows an example of the new class emergence drift that occurs when a new class appears, affecting the ability of the model to adapt to the increased classification task.
- Figure 3(f)) illustrates existing class fusion drift, which occur when the number of classes is reduced by merging existing classes into another.

Concept drifts in terms of speed are characterized by their rate of change and as follows:

- Figure 4(a) shows that abrupt drift (or sudden drift) represents a rapid change in the underlying concept, caused by a sudden change in the data distribution. This is illustrated by an immediate shift from one dominant class to another over the stream.
- Figure 4(b) shows that gradual drift involves a gradual shift from old to new concepts over the stream. This is illustrated by the gradual presence of different classes, indicating a slow shift in the decision boundaries.
- Figure 4(c) shows that incremental drift is a smooth evolution of the concept over time. This is illustrated by a progressive change in class dominance whereby the data distribution and the corresponding concept shift through multiple intermediate states. For example, consider a sensor that gradually degrades.
- Figure 4(d) shows that a recurrent drift occurs when both the previous and the new seen concepts cyclically reappear over time. This is illustrated by the alternating presence of a class, where the underlying concept and data distribution are periodically repeated over the stream.

The presence of noise increases the difficulty of the drift problem, requiring dedicated efforts for drift detection and adaptation. However, in a changing environment, any predictive model should be able to perform the following steps to overcome the concept drift problem [49]:

- Detect and adapt to concept drift (if necessary) and as soon as possible.
- Distinguish between drifts and noise, adapting to changes while being robust to noise.

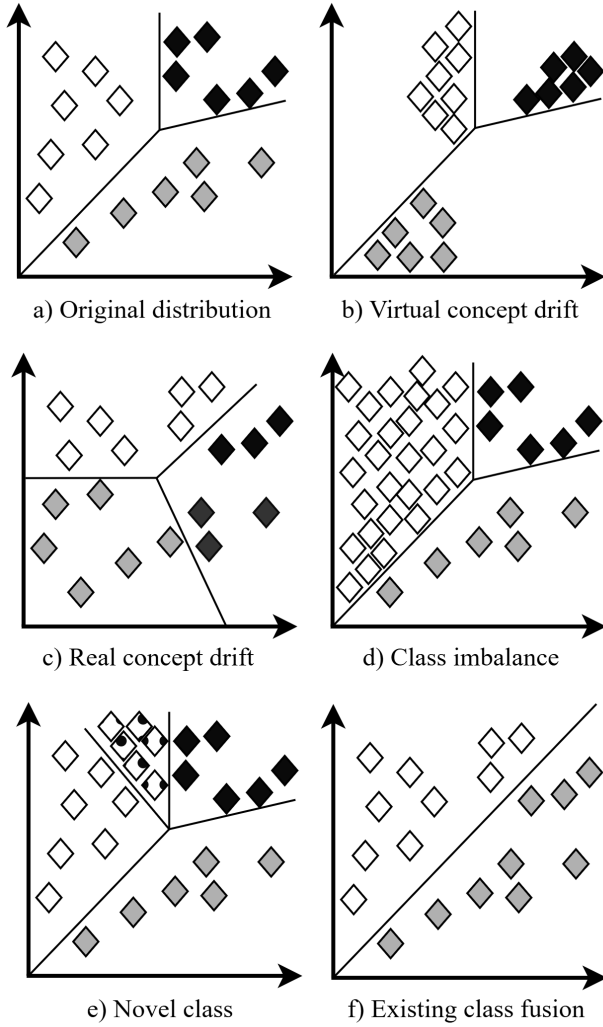


Figure 3. Concept drifts in terms of probability

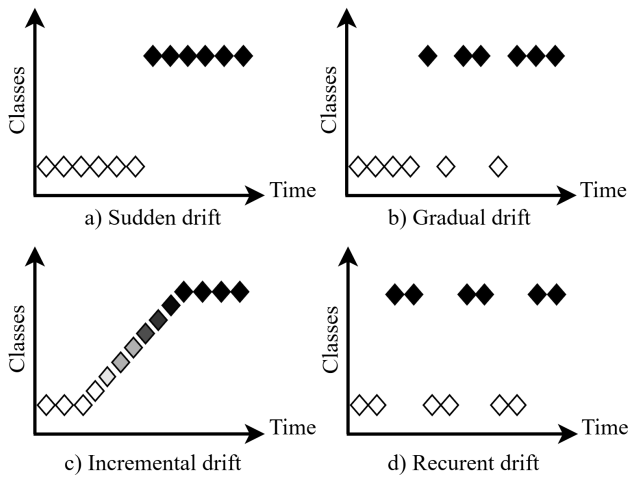


Figure 4. Concept drifts in terms of speed

- Operate faster than the arrival time of the examples and use less or equal to the fixed memory amount for any storage.

B. Catastrophic forgetting

Adapting a trained NN model to new tasks can result in catastrophic forgetting, a phenomenon in which the model forgets previously learned tasks [52]. This issue limits the model's capability to continuously learn. The objective of continuous learning is to learn new concepts effectively without forgetting those already learned. For a given set of labeled data $D_1, D_2, \dots, D_T$ which includes pairs (X, y) of T tasks, the goal is to learn all these tasks in sequence as [53]:

$$\theta^T = \arg \min_{\theta} \sum_{i=1}^T \sum_{(X,y) \in D_i} L(X, y; \theta) \quad (3)$$

where θ^T is adapted from θ^{T-1} , the parameters of the model after training $T-1$ tasks. When learning T , model loses access to data from previously learned T . The aim is to learn the new T from D_T , while maintaining the performance of the prior tasks (from 1 to $T-1$) without any access to $D_1, D_2, \dots, D_{T-1}$. According to [54], for online continual learning of complex audio applications (for example, automatic speech recognition) A model M should, at each time step n :

- Achieve equivalent or better accuracy than the previous model (M_{n-1});
- Achieve better accuracy than continually training the previous model (M_{n-1}) using only the new data D_n ;
- Achieve equivalent accuracy to a model that trained from scratch on the available data at this time $D_{i=0}^n$;
- Require less computational cost than a model that trains from scratch on $D_{i=0}^n$.

Although many proposed approaches can mitigate catastrophic forgetting, each method has only succeeded under certain circumstances, which must be taken into consideration. Achieving human-level performance with a general solution continues to be a challenging goal [55].

C. Overfitting

Overfitting occurs when a DL model is excessively adapted to the specific characteristics and noise of the training data, resulting in a poor generalization to new data. Although numerous studies have successfully designed deep network architectures for ASR such as Very Deep CNN [56] and Deep RNN [57], overfitting remains a critical issue when training DL on popular benchmark datasets [58]. Overfitting is often associated with an overly complex network relative to the task it is intended to perform. The complexity and generalizability of a network depend on its size, typically measured by the number of weights [59], [60], or the magnitude of the weights of the network [61].



To mitigate overfitting, techniques such as regularization, early stopping, dropout, data augmentation, and adopting a simpler model architecture can be applied.

D. Underfitting

Underfitting arises when a DL model fails to adequately capture the underlying patterns of the training data and does not accurately model the target mappings, leading to low accuracy and high loss levels. This issue can result from several factors, including [62], [63]:

- Overly simplistic network design: The network may lack sufficient hidden layers or trainable parameters, rendering it incapable of modeling the complex properties of the data.
- Insufficient training: The network might not have been trained for a sufficient duration, been provided with enough input features, or utilized effective regularization techniques, preventing it from learning the data's features or distributions adequately.

E. Hyperparameter Selection

Hyperparameters are parameters set before the training process which govern various aspects of the learning algorithm. Tuning hyperparameters in stream processing algorithms is a complex and time-consuming task [64]. This complexity is further compounded in non-stationary data stream processing, where data arrive in real-time, and models are expected to make instantaneous predictions as the properties of the data stream evolve. Consequently, hyperparameters that were once effective may become ineffective in the present. Despite considerable efforts in this area, using NNs for data streams requires efficient training and testing, possibly without extensive hyperparameter tuning [65]. Tuning and fine-tuning NN models in non-stationary environments, such as audio processing, can lead to issues like concept drift, overfitting, underfitting, catastrophic forgetting, increased computational resource demands, and higher latency.

F. Biased Learning

Biased learning occurs when the NN model becomes inclined toward predicting the majority class, resulting in high classification accuracy but poor generalization when confronted with balanced or varying data distributions. This problem can be attributed to many factors including [66], [67]:

- Imbalanced data streams
- Inherent network biases

Traditional online learning algorithms are generally designed for balanced data flows; however, most real-world data streams are imbalanced, containing a disproportionate number of samples from one class compared to others [68]. In such cases, high overall accuracy does not necessarily indicate good model performance. Rather, it highlights the

inadequacy of accuracy as the sole metric in imbalanced data scenarios. For example, in [69] the authors used a medical data set with (50 true negative values) as the majority class and (20 true positive values) as the minority class. When they selected half of the data for training and the other half for testing (25 healthy individuals and 10 sick individuals), they found that the accuracy was 90% suggesting that the classification performs well. However, while all negative cases were correctly identified, only 5 out of 10 positive cases were accurately classified, meaning the model incorrectly labeled 5 sick individuals as healthy.

7. CONCLUSIONS AND FUTURE WORK

Sound is a rich source of data, and recent technological advances have enabled new applications for sound analysis that can extract insights from sounds generated by machines, humans, and the natural environment. However, the non-stationary nature of sound, the complexity of feature extraction, the computational cost of audio multi-domain processing, the large volume of audio data, and the complexities of stream processing all pose challenges for real-time audio processing. To enhance real-time responsiveness, fog computing can offload some of the processing to devices on the edge of the network while the remaining data are processed in the cloud. However, ensuring the correctness of the results is highly dependent on the chosen methods, algorithms, and frameworks.

NN have demonstrated outstanding performance in a range of applications including sound/audio processing. However, they face challenges when applied to real-time and streaming data scenarios such as IoST. These challenges include concept drift, catastrophic forgetting, overfitting, underfitting, biased learning, and the complexity of hyperparameter selection. These issues overlap, rendering the application of NN a multi-objective optimization problem. For example, simplifying NN model to prevent overfitting can result in an underfitting problem, and vice versa. Additionally, updating the model with fine-tuning for new tasks can introduce problems such as concept drift, overfitting, underfitting, catastrophic forgetting, increased computational demands, and higher latency. Furthermore, certain NN architectures exhibit biased learning by nature; for instance, standard CNNs focus on high-frequency information in signals.

Due to the interconnected nature of these challenges and the inherent complexity of sound and audio data, our review suggests that current research efforts primarily address specific problems in isolation without offering comprehensive solutions. While some proposed methods effectively mitigate issues such as catastrophic forgetting, developing a universal solution that achieves human-level capability in continual learning essential for non-stationary environments remains an open research question. This study has highlighted several significant challenges and prerequisites for audio stream processing in the context of IoST.

The insights presented in this work may serve as a

reference for future research on developing adaptive and low-latency IoST systems. Future research may explore integrating continual learning methods into cloud and fog layers deployments, and designing lightweight models that handle the concept drift and minimize catastrophic forgetting while maintaining real-time inference.

Furthermore, future IoST systems should support decentralized training paradigms, such as federated or edge collaborative learning. These paradigms reduce bandwidth while enabling models to learn from evolving audio patterns across geographies. Further research efforts should be devoted to optimizing model compression and energy efficiency for dynamic audio environments. Additionally, developing benchmark datasets that simulate real-world acoustic data and non-stationary, multi-source acoustic streams is crucial for testing and validating adaptive IoST systems.

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